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Wafer chuck, temperature control system, and temperature control method

Abstract

A wafer chuck having a holding surface for holding a wafer, the wafer chuck includes: a heating and cooling unit configured to heat or cool the wafer chuck; at least one temperature sensor disposed in the wafer chuck; and a heat flow sensor which is disposed in the wafer chuck, and in which a plurality of thermocouples are connected in series and connecting parts between thermocouples adjacent to each other are alternately disposed at a first depth position from the holding surface and at a second depth position deeper than the first depth position. The heat flow sensor has a plurality of temperature measurement points provided for the respective thermocouples, and the plurality of temperature measurement points are disposed over an entirety of the wafer chuck when the wafer chuck is seen in plan view.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a Continuation of PCT International Application No. PCT/JP2023/020224 filed on May 31, 2023 claiming priorities under 35 U.S.C § 119(a) to Japanese Patent Application No. 2023-016938 filed on Feb. 7, 2023 and Japanese Patent Application No. 2023-076050 filed on May 2, 2023. Each of the above applications is hereby expressly incorporated by reference, in its entirety, into the present application.

BACKGROUND OF THE INVENTION

Field of the Invention

(1) The present invention relates to a wafer chuck, a temperature control system, and a temperature control method.

Description of the Related Art

(2) In a semiconductor manufacturing process, various treatments are applied to a semiconductor wafer to form chips (dice) each including a semiconductor device. The electrical properties of each chip are tested, and then chips are individually divided (diced) by a dicer. Subsequently, each chip is fixed to a lead frame or the like, and is assembled. The electrical properties are tested by a wafer test system that includes a prober and a tester. The prober fixes the wafer on the wafer chuck, and brings a probe into contact with an electrode pad of each chip. The tester provides power supply and various test signals through terminals connected to the probe, analyzes a signal output from the electrode of the chip, and verifies whether a semiconductor device on the chip, which is a test target, normally operates or not.

(3) The prober is required to bring the temperature during testing of the electrical properties to a set temperature, and maintain the temperature constant. Accordingly, to cope with heat generation of the semiconductor device (device to be measured, measurement target device) of a chip to be tested, temperature sensors, such as resistance temperature detectors or thermocouple-type temperature sensors, are installed in a wafer chuck to detect the temperature and the amount of heat generation, and temperature of the wafer chuck is controlled based on the detection results of the temperature sensors.

(4) For example, according to a prober disclosed in Patent Literature 1, temperature sensors are arranged in a wafer chuck (prober chuck), and temperature of the wafer chuck is controlled based on a detection result of a temperature sensor closest to the measurement target device to be tested among the temperature sensors.

CITATION LIST

SUMMARY OF THE INVENTION

(6) In the prior art disclosed in Patent Literature 1, to correctly measure the temperature of the wafer chuck that corresponds to a heat generating part of the device to be measured, it is required to increase the number of temperature sensors installed in the wafer chuck. However, due to a structural limitation in the wafer chuck, the number of installations and the installation positions are limited, and it is difficult to embed a large number of temperature sensors at ideal positions. In the case where a large number of temperature sensors are installed, the number of required sets of signal lines of the temperature sensors increases as increase in the number of temperature sensors, and it is significantly difficult to lay out their wiring.

(7) In particular, in a case where the device to be measured is a SoC (System On Chip) type device, such as a CPU (Central Processing Unit), a GPU (Graphics Processing Unit), or an APU (Accelerated Processing Unit), its heat generation has a locally high heat density in comparison with a memory device. Accordingly, in a case where many temperature sensors cannot be installed in the wafer chuck, it is difficult to appropriately control the temperature of the wafer chuck to cope with heat generation of the device to be measured.

(8) The present invention has been made in view of such situations, and aims to provide a wafer chuck that can measure temperature of the wafer chuck corresponding to a heat generating part of a wafer without increasing the number of installed temperature sensors, and a temperature control system and a temperature control method that can appropriately control the temperature of the wafer chuck.

(9) A wafer chuck to achieve the object of the present invention is a wafer chuck having a holding surface for holding a wafer, the wafer chuck including: a heating and cooling unit configured to heat or cool the wafer chuck; at least one temperature sensor disposed in the wafer chuck; and a heat flow sensor which is disposed in the wafer chuck, and in which a plurality of thermocouples are connected in series and connecting parts between thermocouples adjacent to each other are alternately disposed at a first depth position from the holding surface and at a second depth position deeper than the first depth position, wherein the heat flow sensor has a plurality of temperature measurement points provided for the respective thermocouples, and the plurality of temperature measurement points are disposed over an entirety of the wafer chuck when the wafer chuck is seen in plan view.

(10) According to this wafer chuck, the temperature of the wafer chuck corresponding to the heat generating part of the wafer may be measured without increasing the number of installed temperature sensors.

(11) In the wafer chuck according to another aspect of the present invention, temperature sensors are disposed in a distributed manner. Accordingly, when the device to be measured on the wafer to be tested generates heat, the temperature of the wafer chuck at a part that is not affected by the heat generation may be detected by at least one of the temperature sensors.

(12) A temperature control system to achieve the object of the present invention includes: the above-described wafer chuck; and a temperature control unit configured to calculate a control temperature, based on a detection result of the temperature sensor and on a detection result of the heat flow sensor.

(13) According to the temperature control system, the temperature of the wafer chuck may be appropriately controlled without increasing the number of temperature sensors arranged in the wafer chuck.

(14) In the temperature control system according to another aspect of the present invention, the temperature control unit executes: a process of detecting, based on the detection result of the temperature sensor, a reference temperature that is a temperature at the second depth position in a corresponding part being a part corresponding to a heat generating part of the wafer on the wafer chuck; a process of calculating a vertical temperature difference between the first depth position and

the second depth position in the corresponding part, based on the detection result of the heat flow sensor; and a process of calculating, as the control temperature, a temperature of a heat generating part temperature measurement point that is a temperature of the corresponding part, based on the reference temperature and the vertical temperature difference. Accordingly, the temperature of the wafer chuck may be appropriately controlled without increasing the number of temperature sensors arranged in the wafer chuck.

(15) In the temperature control system according to another aspect of the present invention, the temperature control unit executes: a process of detecting, based on the detection result of the temperature sensor, a reference temperature that is a temperature at the second depth position in a corresponding part being a part corresponding to a heat generating part of the wafer on the wafer chuck; a process of calculating a vertical temperature difference between the first depth position and the second depth position in the corresponding part, based on the detection result of the heat flow sensor; and a process of calculating, as the control temperature, a device temperature that is a temperature of a heat generating part of the wafer, based on the reference temperature, on the vertical temperature difference, and on data that includes physical property values and dimensions of the wafer and the wafer chuck. Accordingly, the temperature of the wafer chuck may be appropriately controlled without increasing the number of temperature sensors arranged in the wafer chuck.

(16) In the temperature control system according to another aspect of the present invention, in the wafer chuck, the plurality of temperature sensors are disposed in a distributed manner, and the temperature control unit detects, as the reference temperature, a lowest temperature among temperatures detected by the plurality of temperature sensors, or an average or a median of the temperatures detected by the plurality of temperature sensors.

(17) In the temperature control system according to another aspect of the present invention, the temperature control unit controls the heating and cooling unit so that the control temperature may be equal to a preset target temperature.

(18) A temperature control method to achieve the object of the present invention is a temperature control method of controlling a temperature of the wafer chuck described above, the method including: a reference temperature detecting step of detecting, by the at least one temperature sensor, a reference temperature of the wafer chuck; a heat flow detecting step of detecting, by the heat flow sensor, a heat flow caused by partial heat generation of the wafer held on the holding surface; and a temperature control step of calculating a control temperature, based on a detection result of the temperature sensor and on a detection result of the heat flow sensor.

(19) In the temperature control method according to another aspect of the present invention, the reference temperature detecting step detects, as the reference temperature, a temperature at the second depth position in a corresponding part that corresponds to a heat generating part of the wafer on the wafer chuck, and the temperature control step includes: a process of calculating a vertical temperature difference between the first depth position and the second depth position in the corresponding part, based on the detection result of the heat flow detecting step; and a process of calculating, as the control temperature, a temperature of a heat generating part temperature measurement point that is a temperature of the corresponding part, based on the reference temperature and the vertical temperature difference.

(20) In the temperature control method according to another aspect of the present invention, the reference temperature detecting step detects, as the reference temperature, a temperature at the second depth position in a corresponding part that corresponds to a heat generating part of the wafer on the wafer chuck, and the temperature control step includes: a process of calculating a vertical temperature difference between the first depth position and the second depth position in the corresponding part, based on the detection result of the heat flow detecting step; and a process of calculating, as the control temperature, a device temperature that is a temperature of a heat generating part of the wafer, based on the reference temperature, on the vertical temperature difference, and on data that includes physical property values and dimensions of the wafer and the wafer chuck.

(21) In the temperature control method according to another aspect of the present invention, the temperature control step includes a process of heating or cooling the wafer chuck so that the control temperature may be a preset target temperature.

(22) According to the present invention, the temperature of the wafer chuck corresponding to the heat generating part of the wafer can be measured without increasing the number of installed temperature sensors. The temperature of the wafer chuck can be appropriately controlled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram showing an entire configuration of a wafer test system.
- (2) FIG. 2 is a schematic configuration diagram of a temperature control system in a first embodiment.
- (3) FIG. 3 shows a schematic plan view and an enlarged sectional view illustrating an internal structure of the wafer chuck.
- (4) FIG. 4 is a functional block diagram showing the configuration of a temperature control device in the first embodiment.
- (5) FIG. 5 is a conceptual diagram of a calculation example of the area of a heat generating part obtained from a heat flow sensor.
- (6) FIG. 6 is a flowchart showing an example of a temperature control method in the temperature control system in the first embodiment.
- (7) FIG. 7 shows a configuration example in a case where the heat flow sensor includes two sensor elements.
- (8) FIG. 8 is a schematic configuration diagram of a temperature control system according to a first modified example.
- (9) FIG. 9 is a schematic configuration diagram of a temperature control system according to a second modified example.
- (10) FIG. 10 is a functional block diagram showing the configuration of a temperature control device in a second embodiment.
- (11) FIG. 11 is an illustration diagram for illustrating a method of calculating a device temperature.

DESCRIPTION OF THE EMBODIMENTS

- (12) Hereinafter, a preferable embodiment of the present invention is described with reference to the accompanying drawings.
- (13) <Wafer Test System>
- (14) FIG. 1 is a schematic diagram showing the entire configuration of a wafer test system 1. Note that description is hereinafter made using an XYZ orthogonal coordinate system where a plane in parallel with the wafer chuck 18 is adopted as an XY plane.
- (15) The wafer test system 1 shown in FIG. 1 includes: a prober 10 that brings a probe 25 into contact with an electrode on each chip on a wafer W; and a tester 30 that is electrically connected to the probe 25, applies current and voltage to each chip for an electric test, and measures properties.
- (16) The prober 10 includes: a base stage 11; a movable base 12 provided on the base stage 11; a Y-axis movable stage 13; an X-axis movable stage 14; a Z-axis movable rotating unit 15; a wafer chuck 18; a wafer alignment camera 19; pillars 20 and 21; a head stage 22; and a probe card 24 attached to the head stage 22.
- (17) The probe card 24 is provided with the probe 25. Note that a needle positioning camera that detects the position of the probe 25, a cleaning mechanism that cleans the probe, and the like are provided. However, their illustration is omitted here.
- (18) The movable base 12, the Y-axis movable stage 13, the X-axis movable stage 14, and the Z-axis movable rotating unit 15 constitute a moving and rotating mechanism that moves the wafer chuck 18 in three-axis directions, and rotates around the Z-axis. Since the moving and rotating mechanism is

widely known, description thereof is omitted here.

(19) The wafer chuck **18** holds the wafer W on which chips are formed, by vacuum suction. A holding surface **18A** for holding the wafer W is provided on the upper surface of the wafer chuck **18**.

(20) In the wafer chuck **18**, a heating and cooling unit **40** as a heating and cooling source is provided so as to perform electrical property tests for chips in a high-temperature state (e.g., 150° C. at the highest) or a low-temperature state (e.g., -40° C. at the lowest). The heating and cooling unit **40** heats or cools the wafer chuck **18**. An appropriate heater or cooler may be applied as the heating and cooling unit **40**. As the heating and cooling unit **40**, various units may be applied that include, for example, a combination of a heater and a cooling plate, a combination of Peltier element and a cooling plate, a double-layer structure having a heating layer made up of a planar heater and a cooling layer provided with a path for a cooling fluid, and a single layer structure of a thermal conductor with an embedded cooling tube around which a heater is wound. The heating and cooling unit **40** is an example of a heating and cooling unit in the present invention.

(21) The wafer chuck **18** is attached on the Z-axis movable rotating unit **15**. The wafer chuck **18** is movable in the three-axis directions (X-, Y-, and Z-axis directions) by the moving and rotating mechanism described above, and is rotatable in a rotation direction (**0** direction) around the Z-axis.

(22) The probe card **24** is disposed above the wafer chuck **18** on which the wafer W is held. The probe card **24** is detachably attached to an opening (probe card attachment part) of the head stage **22** constituting a top board of a casing of the prober **10**.

(23) The probe card **24** includes the probe **25** disposed in conformity with electrode arrangement on the chip to be tested, and is replaced depending on the chip to be tested.

(24) The tester **30** includes a tester main body **31**, and a contact ring **32** provided to the tester main body **31**. The probe card **24** includes electrodes connected to the respective probes **25**. The contact ring **32** includes spring probes disposed so as to contact the respective electrodes. The tester main body **31** is held to face the prober **10** by a support mechanism, not shown.

(25) A control device **90** achieves a function by executing a control program, and controls the operation of the entire prober **10**. For example, the control device **90** controls movement of the aforementioned moving and rotating mechanism (the movable base **12**, the Y-axis movable stage **13**, the X-axis movable stage **14**, and the Z-axis movable rotating unit **15**). The control device **90** controls the operations of the tester **30** and the heating and cooling unit **40**. Note that the control device **90** includes an after-mentioned temperature control device **100** (see FIG. 2).

(26) <Temperature Control System>

(27) Next, a temperature control system incorporated in the prober **10** in this embodiment is described. Note that the temperature control system is an example of a temperature control system in the present invention.

First Embodiment

(28) FIG. 2 is a schematic configuration diagram of the temperature control system **50** in a first embodiment. FIG. 3 is a schematic plan view showing the internal structure of the wafer chuck **18**.

(29) As shown in FIG. 2, the temperature control system **50** in the first embodiment includes the wafer chuck **18**, the temperature control device **100**, temperature sensors **52**, a heat flow sensor **54**, cooling plates **42**, and heaters **44**. The cooling plates **42** and the heaters **44** are components of the heating and cooling unit **40** described above.

(30) The wafer chuck **18** includes a chuck top suction plate **28**. The chuck top suction plate **28** constitutes a part on an upper side (a side where the wafer W is disposed) of the wafer chuck **18**. The chuck top suction plate **28** has a chuck top front surface **28A** that corresponds to the holding surface **18A** of the wafer chuck **18**, and a chuck top back surface **28B** that is opposite to the chuck top front surface **28A** (holding surface **18A**). The cooling plates **42** are disposed in the chuck top suction plate **28**. The heaters **44** are disposed at positions under the chuck top suction plate **28** (opposite to the part on which the wafer W is disposed) and in contact with the chuck top back surface **28B**.

(31) A cooling device **92** is connected to the cooling plates **42**. Under control by the temperature control device **100**, coolant is supplied from the cooling device **92** to the cooling plates **42**. A heater

power source **94** is connected to the heaters **44**. Under control by the temperature control device **100**, heater power is supplied from the heater power source **94** to the heaters **44**.

(32) The temperature sensors **52** are provided in the wafer chuck **18** (chuck top suction plate **28**). Each temperature sensor **52** is made up of, for example, a resistance temperature detector (RTD), a thermocouple (TC) or the like.

(33) The temperature sensors **52** are uniformly (evenly) distributed and disposed (disposed in a distributed manner) in the wafer chuck **18** (chuck top suction plate **28**) in plan view (view in Z-direction) in order to detect an after-mentioned reference temperature T_{ref} of the wafer chuck **18** (see FIG. **3**). In this embodiment, for example, as shown in FIG. **3**, five temperature sensors **52** are provided in the chuck top suction plate **28**. Specifically, one temperature sensor **52** among the five temperature sensors **52** is disposed at the center of the wafer chuck **18**, and the remaining four temperature sensors **52** are uniformly (evenly) disposed along the circumferential direction on the periphery of the wafer chuck **18**. The temperature sensors **52** may be non-uniformly distributed and disposed in the wafer chuck **18** in plan view as long as temperature sensors **52** can detect the reference temperature T_{ref} of the wafer chuck **18**.

(34) Note that “distributed and disposed” means that the temperature sensors **52** are disposed at predetermined intervals, and indicates an arrangement form where when the device to be measured on the wafer **W** to be tested generates heat, at least one temperature sensor **52** among the temperature sensors **52** can detect the temperature of the wafer chuck **18** at a part that is not affected by the heat generation (corresponding to the reference temperature T_{ref} described above). That is, this indicates a state where the temperature sensors **52** are disposed with their distances being separated so that at least one temperature sensor **52** can detect the reference temperature T_{ref} of the wafer chuck **18**, instead of a state where the temperature sensors **52** which are provided according to the number of the devices to be measured on the wafer **W**, are disposed close to respective the devices to be measured.

(35) The heat flow sensor **54** is provided, together with the temperature sensors **52**, in the wafer chuck **18** (chuck top suction plate **28**). The heat flow sensor **54** detects the amount of heat generation of the devices to be measured as a heat flow (heat flux: an amount of heat flowing through a unit area in a unit time period). Note that “heat flow sensor” is also referred to as “heat flux sensor”.

(36) As shown in FIGS. **2** and **3**, the heat flow sensor **54** is formed by connecting thermocouples **56**, as many as the number of temperature measurement points, in series. Output wires **80** are provided at both ends of the heat flow sensor **54**. Each output wires **56** is provided for outputting an output signal (voltage signal) (hereinafter referred as “sensor signal”) indicating the heat flow (heat flux) detected by each heat flow sensor **54**. The output wires **80** are connected to the temperature control device **100**, and is configured such that the sensor signal output from the heat flow sensor **54** may be input into the temperature control device **100**.

(37) Each thermocouple **56** is configured by connecting metals **56A** and **56B** that are of different types, and includes a temperature measurement point **60** serving as a connecting part between the metal **56A** and the **56B**. That is, the heat flow sensor **54** includes the temperature measurement points **60** for the respective thermocouples **56**. The temperature measurement points **60** are respectively disposed at upper positions (positions on a side closer to the chuck top front surface **28A**) in the chuck top suction plate **28**, and are densely disposed over the entire wafer chuck **18** in plan view of the wafer chuck **18** (view in the Z-direction). Note that “densely disposed” indicates an arrangement form where in plan view of the wafer chuck **18**, the temperature measurement points **60** are disposed in a state of having at least a higher arrangement density than the arrangement density of the temperature sensors **52** so as to be capable of detecting the amount of heat, which partially occurs in the wafer **W**, generated in each device to be measured. Preferably, in view of detecting the amount of heat, which partially occurs in the wafer **W**, generated in each device to be measured, the temperature measurement points **60** are uniformly and densely disposed over the entire wafer chuck **18** in plan view of the wafer chuck **18**. However, the disposition is not limited to this example. The temperature measurement points **60** are not necessarily uniformly disposed.

(38) The thermocouples **56** are connected in series, and the heat flow sensor **54** includes reference junctions **62** that are connecting parts between thermocouples **56** adjacent to each other (i.e., the connecting part between the metal **56A** of one thermocouple **56** and the metal **56B** of another thermocouple **56** among every adjacent pair of thermocouples **56**). That is, the heat flow sensor **54** includes multiple reference junctions **62**. Each reference junction **62** is disposed at a predetermined position on the chuck top back surface **28B** which is the lower side of the chuck top suction plate **28**. Note that in this embodiment, for example, each reference junction **62** is disposed at an intermediate position between every adjacent pair of thermocouples **56** connected in series in plan view of the wafer chuck **18** (Z-direction view).

(39) Here, an example of specific attachment form of the heat flow sensor **54** to the wafer chuck **18** is described. As shown in FIG. **3**, which is an enlarged diagram, on the chuck top suction plate **28** included in the wafer chuck **18**, the thermocouple housing bores **34** are provided at positions corresponding to the respective temperature measurement points **60**. Thermocouple housing bores **34** are solid-bottomed elongated bores that are open on the chuck top back surface **28B** of the chuck top suction plate **28**. The distal end parts (parts closer to the corresponding temperature measurement point **60**) of the metal **56A** and the metal **56B**, which constitute the thermocouple **56**, are stored in the corresponding thermocouple housing bore **34**. At the bottom (closer to the chuck top front surface **28A**; upper in FIG. **3**) of each thermocouple housing bore **34**, the distal end parts of the metals **56A** and **56B** constituting the corresponding thermocouple **56** are connected to each other, thus constituting the temperature measurement point **60**.

(40) The base parts of the metals **56A** and **56B** constituting the thermocouples **56** are disposed along the chuck top back surface **28B** outside of the thermocouple housing bores **34**, and are connected to the base parts of the metals **56A** or **56B** of adjacent other thermocouples **56** which are connected in-series. Thus, the base parts of the metals **56A** and **56B** constitute the reference junctions **62**.

(41) According to the form where the thermocouple housing bores **34** are provided in the wafer chuck **18** (chuck top suction plate **28**) as described above, the thermocouples **56**, which are connected in series, may be effectively and easily embedded in the wafer chuck **18**. Note that in this embodiment, the thermocouple housing bores **34** are provided in the wafer chuck **18** is described as a preferable form. However, as long as the temperature measurement points **60** of the heat flow sensor **54** may be densely disposed over the entire wafer chuck **18** in plan view of the wafer chuck **18**, the form of attaching the heat flow sensor **54** to the wafer chuck **18** is not specifically limited.

(42) FIG. **4** is a functional block diagram showing the configuration of the temperature control device **100**. The temperature control device **100** includes a computational circuit made up of various processors, memories and the like. The various processors include a CPU (central processing unit), a GPU (graphics processing unit), an ASIC (application specific integrated circuit), and programmable logic devices [e.g., an SPLD (simple programmable logic devices), a CPLD (complex programmable logic device), and an FPGA (field programmable gate arrays)]. Note that various functions of the temperature control device **100** may be achieved by a single processor, or the same or different types of processors.

(43) The temperature control device **100** executes a control program, not shown, and functions as the reference temperature detecting unit **102**, the temperature difference detecting unit **104**, and the temperature control unit **106**.

(44) The reference temperature detecting unit **102** is connected to the temperature sensors **52**, and obtains the temperature of the wafer chuck **18** detected by each of the temperature sensors **52**. The reference temperature detecting unit **102** detects (selects) the lowest temperature among, or the average or median of the temperatures detected by the respective temperature sensors **52**, as the reference temperature T_{ref} of the wafer chuck **18**. The reference temperature T_{ref} detected by the reference temperature detecting unit **102** is input into the temperature control unit **106**. The reference temperature detecting unit **102** is an example of the reference temperature detecting unit in the present invention.

(45) Note that the “reference temperature T_{ref} ” indicates the temperature of the wafer chuck **18** at a

part that is not affected by heat generation when the device to be measured on the wafer W as a test target generates heat. In this embodiment, the temperature sensor **52** positioned farthest from the device to be measured that is partially generating heat on the wafer W among the temperature sensors **52** is least affected by heat generation of the device to be measured, and detects the lowest temperature. Accordingly, the lowest temperature among the temperatures detected by the respective temperature sensors **52** is selected as the reference temperature T_{ref} . However, in some cases, detection of increase in temperature of the wafer chuck **18** due to heat generation by the devices may be excellent in temperature followability of the wafer chuck **18** in the temperature control. There is also a method of adopting the average, median or the like of the temperatures detected by the respective temperature sensors **52**, as the reference temperature T_{ref} of the wafer chuck **18**.

(46) The reference temperature T_{ref} detected by the reference temperature detecting unit **102** may be regarded as a temperature substantially equal to the temperature on the chuck top back surface **28B** (i.e., a position where the reference junctions **62** of the heat flow sensor **54** are disposed) of the chuck top suction plate **28** at a part of the wafer chuck **18** that is least affected by heat generation by the device to be measured. Thus, the reference temperature T_{ref} is used as a reference temperature for obtaining the temperature T at the heat generating part temperature measurement point from an after-mentioned vertical temperature difference ΔT_d . Note that the “heat generating part temperature measurement point” indicates the temperature measurement point **60** at a position corresponding to the heat generating part of the wafer W among the temperature measurement points **60** constituting the heat flow sensor **54**.

(47) The temperature difference detecting unit **104** is connected to the heat flow sensor **54**, and obtains a sensor signal output from the heat flow sensor **54**. The temperature difference detecting unit **104** extracts an electromotive force (thermal electromotive force) from the sensor signal output from the heat flow sensor **54**, and calculates the vertical temperature difference ΔT_d at the heat generating part temperature measurement point, based on the electromotive force.

(48) The temperature control unit **106** obtains the reference temperature T_{ref} from the reference temperature detecting unit **102**, and obtains the vertical temperature difference ΔT_d at the heat generating part temperature measurement point from the temperature difference detecting unit **104**. The temperature control unit **106** then calculates the temperature T at the heat generating part temperature measurement point, based on the reference temperature T_{ref} and the vertical temperature difference ΔT_d at the heat generating part temperature measurement point. Furthermore, the temperature control unit **106** adopts the calculated temperature T at the heat generating part temperature measurement point as a control temperature (PV value), and controls the heating and cooling unit **40** so that the control temperature may be equal to a preset target temperature (e.g., a test temperature). The temperature control unit **106** is an example of a temperature control unit in the present invention.

(49) <Principle of Temperature Control>

(50) Next, the principle of temperature control in this embodiment is described in detail.

(51) At the heat flow sensor **54** according to this embodiment, a temperature difference between the front and back of the heat flow sensor **54** occurs when a heat flow passes in the thickness direction (Z-direction) of the wafer chuck **18** through the heat flow sensor **54** due to heat generation of the device to be measured on the wafer W. That is, when the heat flow passes through the heat flow sensor **54** in the thickness direction of the wafer chuck **18**, a temperature difference occurs between one side (the chuck top front surface **28A** side) and the other side (chuck top back surface **28B** side) of the wafer chuck **18**. Accordingly, an electromotive force occurs between the one side and the other side of the wafer chuck **18** by the Seebeck effect. The heat flow sensor **54** outputs, as the sensor signal, the electromotive force (thermal electromotive force) caused based on the heat flow flowing between the front and back of the heat flow sensor **54**.

(52) Because of the configuration described above, the heat flow sensor **54** in this embodiment may calculate the temperature difference between the front and back of the heat flow sensor **54**, based on the sensor signal output by the heat flow sensor **54**, i.e., the electromotive force at the heat flow

sensor 54.

(53) Here, when it is assumed that the electromotive force of the heat flow sensor 54 is V, and the heat flow flowing through the heat flow sensor 54 is q, a relationship of $q = \alpha \cdot \text{Math.} V$ (α is a sensor constant) holds. When it is assumed that the temperature difference between the front and back of the heat flow sensor 54 is ΔT , the heat flow q passing through the heat flow sensor 54 is proportional to the temperature difference ΔT at the heat flow sensor 54. Consequently, there is a correlation between the electromotive force V at the heat flow sensor 54, and the temperature difference ΔT at the heat flow sensor 54. Accordingly, when data indicating their relationship (conversion data between the electromotive force and the temperature difference) are obtained in advance through experiment or design, the temperature difference ΔT may be obtained from the electromotive force at the heat flow sensor 54.

(54) When it is assumed that the temperature difference obtained from the electromotive force at the heat flow sensor 54 is ΔT [$^{\circ}\text{C.}$], the rate of heat passing through the heat flow sensor 54 is Q [W], the area of the heat flow sensor 54 is S [m.sup.2], the heat flow distance of the heat flow sensor 54 (the distance between the front and back of the heat flow sensor 54) is Lj [m], and the thermal conductivity of the material in which the heat flow sensor 54 is embedded is k [W/mK], the heat rate Q passing through the heat flow sensor 54 is obtained by the following Expression (1).

[Formula1]

(55)
$$Q = -k \cdot \text{Math.} S_{\frac{T}{Lj}} \quad \text{Expression(1)}$$

(56) When it is assumed that the heat rate at the heat generating part is Qd [W], and the area (heat generating part area) of the heat generating part is Sd [m.sup.2], the heat rate Qd at the heat generating part is as represented by the following Expression 2.

[Formula2]

(57)
$$Qd = Q \frac{Sd}{S} = -k \cdot \text{Math.} S_{\frac{T}{Lj}} \cdot \text{Math.} \frac{Sd}{S} = -k \cdot \text{Math.} Sd_{\frac{T}{Lj}} \quad \text{Expression(2)}$$

(58) In other words, the heat rate Qd at the heat generating part has a value obtained by multiplying the heat rate Q passing through the heat flow sensor 54 by the area ratio (Sd/S) of the heat generating part area Sd to the area S of the heat flow sensor 54. Furthermore, when the heat rate Q passing through the heat flow sensor 54 is replaced using the relational expression represented by Expression (1), heat rate Qd at the heat generating part may be represented by the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54.

(59) Further, when it is assumed that the vertical temperature difference at the heat generating part temperature measurement point is ΔTd [$^{\circ}\text{C.}$], the vertical temperature difference ΔTd at the heat generating part temperature measurement point is as represented by the following Expression (3).

[Formula3]

(60)
$$Td = \frac{Qd \cdot \text{Math.} Lj}{-k \cdot \text{Math.} Sd} = \frac{-k \cdot \text{Math.} S_{\frac{T}{Lj}} \cdot \text{Math.} Lj}{-k \cdot \text{Math.} Sd} = T \frac{S}{Sd} \quad \text{Expression(3)}$$

(61) In other words, the vertical temperature difference ΔTd at the heat generating part temperature measurement point has a relationship that is proportional to the heat rate Qd at the heat generating part and to the heat flow distance Lj, and is inversely proportional to the heat generating part area Sd. When the heat rate Qd at the heat generating part is replaced using the relational expression represented by Expression (2), the vertical temperature difference ΔTd at the heat generating part temperature measurement point is a value obtained by multiplying the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54 by the reciprocal of the area ratio described above (i.e., the area ratio of the area S of the heat flow sensor 54 to the heat generating part area Sd).

(62) As described above, the vertical temperature difference ΔTd at the heat generating part temperature measurement point is the temperature difference obtained by converting the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54 using the heat generating part area Sd. Accordingly, when it is assumed that the temperature at the heat generating

part temperature measurement point is T [$^{\circ}\text{C}$.], the temperature T at the heat generating part temperature measurement point is a value obtained by adding the vertical temperature difference ΔT_d at the heat generating part temperature measurement point to the reference temperature T_{ref} (the lowest temperature among the temperatures detected by the respective temperature sensors 52), as represented by the following Expression (4).

[Formula4]

$$(63) \quad T = T_{\text{ref}} + \Delta T_d \quad \text{Expression(4)}$$

(64) Note that the reference temperature T_{ref} is the reference temperature at a part (i.e., the back side (chuck top back surface 28B) of the heat flow sensor 54) which is the reference of the vertical temperature difference ΔT_d at the heat generating part temperature measurement point. The temperature sensor 52 at a position farthest from the device to be measured that is partially generating heat in the wafer W, detects the lowest temperature. Accordingly, the temperature detected by this temperature sensor 52 may be assumed to be substantially equal to the temperature on the back side (chuck top back surface 28B) of the heat flow sensor 54. Consequently, as shown in the Expression (4), the temperature T at the heat generating part temperature measurement point may be obtained by adding the vertical temperature difference ΔT_d at the heat generating part temperature measurement point to the reference temperature T_{ref} detected by the reference temperature detecting unit 102.

(65) <Concept of Calculation Example of Temperature of Heat Generating Part>

(66) FIG. 5 is a conceptual diagram of a calculation example of the area of the heat generating part obtained from the heat flow sensor 54. Note that the calculation example described below assumes the following conditions as an example. The heat flow distance of the heat flow sensor 54: $L_j=0.01$ [m] The thermal conductivity of the material in which the heat flow sensor 54 is embedded: $k=180$ [W/mK].

[In a Case where there is One Heat Generating Part]

(67) Reference numeral 5A in FIG. 5 designates an example of the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54 in a case where the number of heat generating parts HP is one in the wafer W. It is herein assumed that the heat rate of the heat generating part HP is $Q_d=100$ [W], and the area of the heat generating part is $S_d=0.000625$ [m.sup.2] (25 mm×25 mm)

(68) In the example designated by reference numeral 5A in FIG. 5, the electromotive force at the heat flow sensor 54 is 0.353 [mV], and the temperature difference obtained from the electromotive force is $\Delta T=8.89$ [$^{\circ}\text{C}$.].

(69) Meanwhile, an actual vertical temperature difference ΔT_d is obtained from the heat rate Q_d of the heat generating part HP using the leftmost relational expression in Expression (3), and the following Expression (5) holds. Note that the sign of the vertical temperature difference ΔT_d at the heat generating part temperature measurement point in this case is determined depending on the direction of the heat flow passing through the heat flow sensor 54.

[Formula5]

$$(70) \quad T_d = \frac{Q_d \cdot \text{Math. } L_j}{-k \cdot \text{Math. } S_d} = \frac{100 \times 0.01}{-180 \times 0.000625} = \frac{1}{-0.1125} = -8.889[^{\circ}\text{C}] \quad \text{Expression(5)}$$

(71) In the case of the example designated by 5A in FIG. 5, the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54 is equal to the absolute value of the vertical temperature difference ΔT_d obtained from the heat rate Q_d of the heat generating part HP (i.e., $\Delta T=|\Delta T_d|$).

(72) [In a Case where there are Two Heat Generating Parts]

(73) Reference numeral 5B in FIG. 5 designates an example of the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54 in a case where the number of heat generating parts HP is two in the wafer (a case of a heat generating area twice as large as the heat generating area of the heat generating part HP shown in reference numeral 5A in FIG. 5). It is herein

assumed that the heat generating part HP is $Q_d=200$ [W], and the area of the heat generating part is $S_d=0.00125$ [m.sup.2] (25 mm×50 mm)

(74) In the example designated by reference numeral 5B in FIG. 5, the electromotive force at the heat flow sensor 54 is 0.706 [mV], and the temperature difference obtained from the electromotive force V is $\Delta T=17.78$ [° C.].

(75) Meanwhile, the vertical temperature difference ΔT_d at the heat generating part temperature measurement point is obtained from the heat rate Q_d of the heat generating part HP using the leftmost relational expression in Expression (3), and the following Expression (6) holds. Note that the sign of the vertical temperature difference ΔT_d at the heat generating part temperature measurement point in this case is determined depending on the direction of the heat flow passing through the heat flow sensor 54.

[Formula6]

$$(76) \quad T_d = \frac{Q_d \cdot \text{Math. Lj}}{-k \cdot \text{Math. Sd}} = \frac{200 \times 0.01}{-180 \times 0.00125} = \frac{2}{-0.225} = -8.889[\text{°C}] \quad \text{Expression(6)}$$

(77) In the example designated by reference numeral 5B in FIG. 5, the heat generating area of the heat generating part HP in the wafer W is twice as large as that in the case of the example designated by reference numeral 5A in FIG. 5. Consequently, the electromotive force at the heat flow sensor 54 increases proportionally to the heat generating area of the heat generating part HP, so as to be twice accordingly. Thus, according to comparison between absolute values of the temperature differences, the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54 becomes twice as large as the vertical temperature difference ΔT_d at the heat generating part temperature measurement point obtained by the heat rate Q_d of the heat generating part HP.

(78) As described above, the electromotive force at the heat flow sensor 54 changes according to the change in area of the heat generating part HP in the wafer W, and according to the change, the temperature difference ΔT obtained from the electromotive force also changes. That is, the temperature difference ΔT obtained from the electromotive force at the heat flow sensor 54 is an apparent temperature difference that changes according to change in area of the heat generating part HP, and deviates from the actual temperature difference. Accordingly, in this embodiment, a value obtained by converting the temperature difference ΔT calculated from the electromotive force at the heat flow sensor 54 using the heat generating part area S_d is adopted as the vertical temperature difference ΔT_d at the heat generating part temperature measurement point, so as to obtain the temperature difference that is substantially equal to the actual temperature difference at the heat generating part temperature measurement point.

(79) <Method of Obtaining Actual Control Temperature>

(80) Next, a method of obtaining the actual control temperature (the vertical temperature difference ΔT_d at the heat generation temperature measurement point) performed by the temperature control device 100 in the embodiment is described.

(81) While the principle of temperature control in this embodiment is as described above, the output value (electromotive force) of the actual heat flow sensor 54 is determined based on the number of thermocouples 56 constituting the heat flow sensor 54 and their mounting density (i.e., the area per pair of thermocouples 56). Accordingly, in this embodiment, the area S in which the heat flow sensor 54 is embedded is calculated as an area S_s per pair of thermocouples 56, thus obtaining the vertical temperature difference ΔT_d at the heat generating part temperature measurement point in the expressions (1) to (3) described above. Note that the calculation example of the vertical temperature difference ΔT_d at the heat generating part temperature measurement point described below is an example, and assumes the following conditions. The area ($\phi 300$ mm) in which the heat flow sensor is embedded: $S_d=0.070875$ [m.sup.2] The number of temperature measurement points of the heat flow sensor: $N_s=114$ [points] The area per pair of thermocouple:

$$S_s = S_d \div N_s = 0.070875 \div 114 = 0.000622 \text{ [m.sup.2]} = S$$

[In a Case where there is One Heat Generating Site]

(82) In the example designated by reference numeral 5A in FIG. 5, provided that the area of the heat

generating part is $S_d=0.000625$ [m.sup.2] (25 mm×25 mm), the electromotive force at the heat flow sensor **54** is 0.353 [mV], and the temperature difference obtained from the electromotive force is $\Delta T=8.89$ [° C.]. The vertical temperature difference ΔT_d at the heat generating part temperature measurement point is obtained by converting the temperature difference ΔT obtained from the electromotive force at the heat flow sensor **54**, using the heat generating part area S_d , as represented by the following Expression (7).

[Formula7]

$$(83) \quad T_d = T \frac{S_d}{S_{sd}} = 8.89 \frac{0.000622}{0.000625} = 8.89 \times 0.9952 = 8.847[^\circ\text{C}] \quad \text{Expression(7)}$$

[In Case where there are Two Heat Generating Sites]

(84) In the example designated by reference numeral **5B** in FIG. 5, it is assumed that the area of the heat generating part is $S_d=0.00125$ [m.sup.2] (25 mm×50 mm), the electromotive force at the heat flow sensor **54** is 0.706 [mV], and the temperature difference obtained from the electromotive force is $\Delta T=17.78$ [° C.]. The vertical temperature difference ΔT_d at the heat generating part temperature measurement point is obtained by converting the temperature difference ΔT obtained from the electromotive force at the heat flow sensor **54**, using the heat generating part area S_d , as represented by the following Expression (8).

[Formula8]

$$(85) \quad T_d = T \frac{S_d}{S_{sd}} = 17.78 \frac{0.000622}{0.00125} = 17.78 \times 0.4976 = 8.847[^\circ\text{C}] \quad \text{Expression(8)}$$

(86) As described above, in the temperature control device **100** in this embodiment, even if the area of the heat generating part HP changes, the vertical temperature difference ΔT_d at the heat generating part temperature measurement point (actual temperature difference) may be calculated from the temperature difference ΔT (apparent temperature difference) obtained from the electromotive force at the heat flow sensor **54**. In controlling the heating and cooling unit **40**, the value obtained by adding the vertical temperature difference ΔT_d at the heat generating part temperature measurement point to the reference temperature T_{ref} (the temperature substantially corresponding to the temperature on the back of the heat flow sensor **54**) detected by the reference temperature detecting unit **102** is adopted as the temperature T at the heat generating part temperature measurement point, the temperature T is adopted as the control temperature (PV value), so that the temperature of the wafer chuck **18** may be appropriately controlled depending on the heat generation state of the device to be measured on the wafer W .

(87) <Temperature Control Method>

(88) Next, a temperature control method in the temperature control system **50** in the first embodiment is described. FIG. 6 is a flowchart showing an example of the temperature control method in the temperature control system **50** in the first embodiment.

(89) First, the wafer W is held on the wafer chuck **18**, alignment between the wafer W and the probe card **24** is achieved, and subsequently, an electrical property test of the wafer W is started. From start to end of the electrical property test of the wafer W , the flowchart shown in FIG. 6 is executed.

(90) After the flowchart in FIG. 6 is started, first, a set temperature determining step (step **S10**) is performed. In the set temperature determining step, the temperature control unit **106** determines the set temperature of the wafer chuck **18**. The set temperature of the wafer chuck **18** is a target temperature (e.g., a test temperature) when the temperature control unit **106** executes the temperature control for the wafer chuck **18**. Note that the set temperature of the wafer chuck **18** is not necessarily equal to the test temperature, and may be, for example, the set temperature may be set based on the test temperature.

(91) After the set temperature determining step is performed, a reference temperature detecting step (step **S12**) and a vertical temperature difference detecting step (step **S14**) are performed in parallel. The reference temperature detecting step and the vertical temperature difference detecting step are not necessarily executed at the same timing (at the same time). However, in a control temperature determining step (step **S16**) that is subsequently performed, the control temperature is determined

based on the detection results in both the steps. Accordingly, it is preferable that the reference temperature detecting step and the vertical temperature difference detecting step be executed at the same timing.

(92) In the reference temperature detecting step (step **S12**), the temperatures detected by the respective temperature sensors **52** are input into the reference temperature detecting unit **102**. The reference temperature detecting unit **102** detects (select) the lowest temperature among the temperatures detected by the respective temperature sensors **52**, as the reference temperature T_{ref} of the wafer chuck **18**. The reference temperature T_{ref} of the wafer chuck **18** is a temperature serving as a reference when the temperature T at the heat generating part temperature measurement point is obtained from the after-mentioned vertical temperature difference ΔT_d at the heat generating part temperature measurement point, and is a temperature that may be assumed as a temperature substantially equal to the temperature on the back side (the chuck top back surface **28B** of the chuck top suction plate **28**) of the heat flow sensor **54**. The reference temperature T_{ref} detected by the reference temperature detecting step is input into the temperature control unit **106**.

(93) In the vertical temperature difference detecting step (step **S14**), the heat flow caused by partial heat generation in the wafer **W** held on the wafer chuck **18** is detected by the heat flow sensor **54**, the sensor signal (electromotive force) output from the heat flow sensor **54** is input into the temperature difference detecting unit **104**. The temperature difference detecting unit **104** obtains the temperature difference ΔT between the front side and back side of the heat flow sensor **54** from the electromotive force at the heat flow sensor **54**. The temperature difference detecting unit **104** then calculates a value obtained by converting the temperature difference ΔT obtained from the electromotive force at the heat flow sensor **54**, using the heat generating part area S_d , as the vertical temperature difference ΔT_d at the heat generating part temperature measurement point. Here, the heat generating part area S_d is obtained by referring to test information (including the area of the device to be measured) stored in a memory unit (not shown). Since the method of calculating the vertical temperature difference ΔT_d at the heat generating part temperature measurement point is as described above, the detailed description thereof is omitted here.

(94) After the reference temperature detecting step and the vertical temperature difference detecting step are performed, a temperature control step (step **S16**) is performed. In the temperature control step, the temperature control unit **106** calculates the temperature T at the heat generating part temperature measurement point, based on the reference temperature T_{ref} input from the reference temperature detecting unit **102** and on the vertical temperature difference ΔT_d at the heat generating part temperature measurement point input from the temperature difference detecting unit **104**. Specifically, the temperature T at the heat generating part temperature measurement point is obtained as a value obtained by adding the vertical temperature difference ΔT_d to the reference temperature T_{ref} (i.e., $T = T_{ref} + \Delta T_d$).

(95) The temperature control unit **106** adopts the thus calculated temperature T at the heat generating part temperature measurement point, as the control temperature (PV value), and controls the heating and cooling unit **40** so that the temperature T at the heat generating part temperature measurement point serving as the control temperature may approach the set temperature of the wafer chuck **18** determined in the set temperature determining step. For example, if the temperature T at the heat generating part temperature measurement point is lower than the set temperature, the heater power is supplied to the heaters **44** from the heater power source **94** according to control by the temperature control unit **106**, and the heaters **44** heat the wafer chuck **18**. If the temperature T at the heat generating part temperature measurement point is higher than the set temperature, the coolant is supplied from the cooling device **92** to the cooling plates **42** according to control by the temperature control unit **106**, and the cooling plates **42** cool the wafer chuck **18**. Accordingly, depending on the heat generation state of the device to be measured on the wafer **W**, temperature control (heating and cooling control) for the wafer chuck **18** is performed so that the temperature of the wafer chuck **18** (temperature T at the heat generating part temperature measurement point) can approach the set temperature.

(96) After the temperature control step is performed as described above, a determining step (step **S18**) is performed. In the determining step, the temperature control unit **106** determines whether the electrical property test for the wafer **W** has been finished or not. If the temperature control unit **106** determines that the electrical property test for the wafer **W** has not been finished in the determining step (case of determination result of No), the processes from step **S12** to step **S18** are repeated. On the other hand, if the temperature control unit **106** determines that the electrical property test for the wafer **W** has been finished (case of determination result of Yes), the control by the temperature control unit **106** is finished. Thus, this flowchart is finished.

(97) As described above, in the temperature control system **50** according to the embodiment, both the temperature sensors **52** and the heat flow sensor **54** which are disposed by connecting the thermocouples **56** in series (the temperature measurement points **60** are densely disposed over the entire wafer chuck **18**), are embedded in the wafer chuck **18**. The temperature control system **50** calculates the temperature **T** at the heat generating part temperature measurement point (the temperature of the wafer chuck **18** corresponding to the heat generating part of the wafer **W**), based on the detection results of the temperature sensors **52** and on the detection result of the heat flow sensor **54**, and performs the temperature control (heating and cooling control) for the wafer chuck **18** so that the temperature **T** may be equal to a preliminarily set temperature. Accordingly, the temperature of the wafer chuck **18** may be appropriately controlled without increasing the number of temperature sensors **52** arranged in the wafer chuck **18**.

(98) Note that in the embodiment, the form where the five temperature sensors **52** are disposed in the wafer chuck **18** (chuck top suction plate **28**) is described as a preferable form. The number of temperature sensors **52** disposed in the wafer chuck **18** is not limited so long as the number of temperature sensors **52** is sufficiently smaller than the number of devices to be measured on the wafer **W**, and allows detection of the reference temperature **Tref** of the wafer chuck **18**. For example, a form where two to four temperature sensors **52** are disposed, or a form where six or more are disposed may be adopted. As long as the reference temperature **Tref** of the wafer chuck **18** may be detected, a form where a single temperature sensor **52** is disposed may be adopted. That is, to detect the reference temperature **Tref** of the wafer chuck **18**, at least one temperature sensor **52** is required to be disposed in the wafer chuck **18**.

(99) In this embodiment, the case where the heat flow sensor **54** is made up of one sensor element (an integrated object where the thermocouples **56** are connected in series) is described as a preferable form. There is no limitation to this. The heat flow sensor **54** may be made up of multiple sensor elements.

(100) FIG. 7 shows a configuration example where the heat flow sensor **54** is made up of two sensor elements. FIG. 7 is a schematic plan view showing the internal structure of the wafer chuck **18**. Note that FIG. 7 shows the configuration example where the heat flow sensor **54** is made up of two sensor elements, as an example. It is however a matter of course that the sensor may be made up of three or more sensor elements.

(101) In the configuration example shown in FIG. 7, the heat flow sensor **54** is made up of two sensor elements **54A** and **54B**. Specifically, in plan view of the wafer chuck **18** (Z-direction view), the sensor element **54A** is disposed in a region on one side (the left side in FIG. 7), and the sensor element **54B** is disposed on the other side (the right side in FIG. 7) of the wafer chuck **18**.

(102) The two sensor elements **54A** and **54B** have a configuration basically similar to the configuration of the heat flow sensor **54** in the embodiment described above. The two sensor elements **54A** and **54B** are each configured to include thermocouples **56** connected in series, and each have temperature measurement points **60** densely disposed over the entire corresponding arrangement region.

(103) The both ends of the two sensor elements **54A** and **54B** are respectively provided with output wires **80A** and output wires **80B** for outputting sensor signals to the temperature control device **100**. Each of the output wires **80A** and output wires **80B** are connected to the temperature control device **100**. It is configured such that the sensor signals output from the respective sensor elements **54A** and

54B may be input into the temperature control device **100**.

(104) The temperature control device **100** obtains the vertical temperature difference ΔT_d at the heat generating part temperature measurement point, based on the sensor signal (electromotive force) output from the sensor element disposed in the region where the device to be measured resides, between the two sensor elements **54A** and **54B**. Since the method of calculating the vertical temperature difference ΔT_d at the heat generating part temperature measurement point is substantially similar to that in this embodiment, the detailed description thereof is omitted here.

(105) In this embodiment, the configuration is described where: the cooling plates **42** are disposed as the heating and cooling unit **40** in the chuck top suction plate **28**; and the heaters **44** are disposed on the back side (the opposite side of the side on which the wafer **W** is disposed) of the chuck top suction plate **28**. However, the configuration is not limited to this. For example, other configuration examples described later may be adopted.

(106) FIG. **8** is a schematic configuration diagram of a temperature control system **50A** according to a first modified example. Note that parts common to those in the first embodiment are assigned the same symbols, and the description thereof is omitted.

(107) As shown in FIG. **8**, the temperature control system **50A** according to the first modified example includes cooling plates **42**, and a Peltier element **46**. The cooling plates **42** and the Peltier element **46** are components of the heating and cooling unit **40**, and are provided in the wafer chuck **18**. Specifically, the Peltier element **46** is disposed in a state of being in contact with the back side (chuck top back surface **28B**) of the chuck top suction plate **28**. The cooling plates **42** are disposed on the opposite side of the chuck top suction plate **28**, with the Peltier element **46** intervening therebetween. In other words, the Peltier element **46** is disposed between the chuck top suction plate **28** and the cooling plates **42**.

(108) A Peltier power source **96** is connected to the Peltier element **46**. Under control by the temperature control device **100**, Peltier power is supplied from the Peltier power source **96** to the Peltier element **46**. The cooling device **92** is connected to the cooling plates **42**. Under control by the temperature control device **100**, coolant is supplied from the cooling device **92** to the cooling plates **42**.

(109) According to the first modified example, control is achieved with combination of heating by the Peltier element **46** and cooling by the cooling plates **42**, thus allowing temperature control for the wafer chuck **18** in a manner similar to that of the first embodiment. Note that cooling by the Peltier element **46** may be further incorporated to the configuration so as to perform control.

(110) FIG. **9** is a schematic configuration diagram of a temperature control system **50B** according to a second modified example. Note that parts common to those in the first embodiment are assigned the same symbols, and the description thereof is omitted.

(111) As shown in FIG. **9**, the temperature control system **50B** according to the second modified example is common to that in the first embodiment in that the cooling plates **42** and the heaters **44** are provided in the wafer chuck **18**. However, the arrangement form between the cooling plates **42** and the heaters **44** is different from that in the first embodiment. Specifically, the heater **44** is disposed in a state of being in contact with the back side (chuck top back surface **28B**) of the chuck top suction plate **28**. The cooling plates **42** are disposed on the opposite side of the chuck top suction plate **28**, with the heaters **44** intervening therebetween.

(112) Also according to the second modified example, control is achieved with combination of heating by the heaters **44** and cooling by the cooling plates **42**, thus allowing temperature control for the wafer chuck **18** in a manner similar to that in the first embodiment.

Second Embodiment

(113) Next, a temperature control system **50** in a second embodiment is described. In the first embodiment described above, based on the detection results of the temperature sensors **52** and the detection result of the heat flow sensor **54**, the temperature **T** at the heat generating part temperature measurement point (the temperature of the wafer chuck **18** corresponding to the heat generating part of the wafer **W**) is calculated, and temperature control for the wafer chuck **18** is performed based on

the calculated temperature T at the heat generating part temperature measurement point. Meanwhile, according to the second embodiment, based on the detection results of the temperature sensors **52**, on the detection result of the heat flow sensor **54**, and on information about physical property values, dimensions and the like from the heat generating part temperature measurement point to the heat generating part, the device temperature corresponding to the temperature of the heat generating part of the wafer W is calculated, and based on the calculated device temperature, the temperature control for the wafer chuck **18** is performed.

(114) Note that the second embodiment has the basically same configuration as the first embodiment described above except that the control temperature when the temperature control is performed for the wafer chuck **18** is the device temperature. Hereinafter, parts common to those in the first embodiment are assigned the same symbols, and the description thereof is omitted.

(115) FIG. **10** is a block diagram of a temperature control device **100A** in the second embodiment. As shown in FIG. **10**, the temperature control device **100A** in the second embodiment includes a temperature control unit **106A** that has a device temperature calculating function.

(116) As in the first embodiment described above, the reference temperature T_{ref} detected by the reference temperature detecting unit **102**, and the vertical temperature difference ΔT_d at the heat generating part temperature measurement point detected by the temperature difference detecting unit **104** are input into the temperature control unit **106A**. Information about the physical property values, dimensions and the like from the heat generating part temperature measurement point to the heat generating part (i.e., the wafer chuck **18** and the wafer W) is preliminarily registered, as device temperature calculation condition data for calculating the device temperature, in a memory unit (not shown). The temperature control unit **106A** may obtain the device temperature calculation condition data from the memory unit.

(117) The temperature control unit **106A** calculates the temperature of the heat generating part of the wafer W (device temperature), based on the reference temperature T_{ref} obtained from the reference temperature detecting unit **102**, on the vertical temperature difference ΔT_d at the heat generating part temperature measurement point obtained from the temperature difference detecting unit **104**, and on the device temperature calculation condition data obtained from the memory unit. The temperature control unit **106A** then adopts the device temperature as the control temperature, and executes temperature control (heating and cooling control) for the wafer chuck **18** so that the device temperature may approach the preliminarily set temperature (e.g., a test temperature).

(118) Here, a method of calculating the device temperature performed in the second embodiment is described. FIG. **11** is an illustration diagram for illustrating the method of calculating the device temperature. Here, the thermal conductivity K_w of the wafer W , the thermal conductivity K_j of the material in which the heat flow sensor **54** is embedded, the distance L_c between the temperature measurement points **60** of the heat flow sensor **54** and the holding surface **18A** of the wafer chuck **18**, the distance L_w between the back of the wafer W and the heat generating part HP and the like are preliminarily registered, as the device temperature calculation condition data, in the memory unit.

(119) As shown in FIG. **11**, when it is assumed that the temperature difference between the temperature measurement points **60** of the heat flow sensor **54** and the holding surface **18A** of the wafer chuck **18** is ΔT_c , the temperature difference between the back of the wafer W and the heat generating part HP is ΔT_w , and the temperature of the heat generating part HP of the wafer W is T_w , the device temperature T_w can be obtained as represented by the following Expression (9).

[Formula9]

(120)
$$T_w = T_{ref} + \Delta T_d + \Delta T_c + T_w \quad \text{Expression(9)}$$

(121) Note that the reference temperature T_{ref} , and the vertical temperature difference ΔT_d at the heat generating part temperature measurement point are respectively detected by the reference temperature detecting unit **102** and the temperature difference detecting unit **104** in a manner similar to that in the first embodiment.

(122) When it is assumed that the heat rate at the heat generating part HP is Q_d [W], the temperature

difference ΔT_c between the temperature measurement points **60** of the heat flow sensor **54** and the holding surface **18A** of the wafer chuck **18**, and the temperature difference ΔT_w between the back of the wafer **W** and the heat generating part **HP** are respectively obtained by the following Expressions (10) and (11). Here, S_d is the area of the heat generating part **HP** of the wafer **W** (area of the heat generating part).

$$(123) \quad T_c = \frac{Q_d \cdot \text{Math. } L_c}{-K_j \cdot \text{Math. } S_d} \quad \text{Expression(10)} \quad T_w = \frac{Q_d \cdot \text{Math. } L_w}{-K_w \cdot \text{Math. } S_d} \quad \text{Expression(11)}$$

(124) The heat rate Q_d at the heat generating part **HP** may be obtained using the leftmost relational expression in Expression (3) described above. That is, the heat rate Q_d at the heat generating part **HP** may be obtained as represented by the following Expression (12), using the vertical temperature difference ΔT_d at the heat generating part temperature measurement point (i.e., the value obtained by converting the temperature difference ΔT obtained from the electromotive force at the heat flow sensor **54** using the area S_d of the heat generating part). Here, L_j is the heat flow distance of the heat flow sensor **54** (the distance between the front side and back side of the heat flow sensor **54**).

$$(125) \quad Q_d = \frac{T_d \cdot (-K_j \cdot \text{Math. } S_d)}{L_j} \quad \text{Expression(12)}$$

(126) Consequently, through use of Expressions (9) to (12), the device temperature T_w may be calculated.

(127) (Calculation Example of Device Temperature)

(128) Next, a calculation example of the device temperature T_w is described. Note that the calculation example of the device temperature T_w described here assumes the following conditions as an example. The reference temperature of the wafer chuck **18**: $T_{ref}=100[^\circ \text{C}]$ The heat flow distance of the heat flow sensor **54**: $L_j=0.01 \text{ [m]}$ The distance between the temperature measurement points **60** of the heat flow sensor **54** and the holding surface **18A** of the wafer chuck **18**: $L_c=0.005 \text{ [m]}$ The distance between the back surface of the wafer **W** and the heat generating part **HP**: $L_w=0.0005 \text{ [m]}$ The thermal conductivity of the material in which the heat flow sensor **54** is embedded: $K_j=180 \text{ [W/mK]}$ The thermal conductivity of the material of the wafer **W**: $K_w=160 \text{ [W/mK]}$ The area of the heat generating part: $S_d=0.000625 \text{ [m.sup.2]}$ The temperature difference obtained from the electromotive force V at the heat flow sensor **54**: $\Delta T=8.89[^\circ \text{C}]$ The number of temperature measurement points of the heat flow sensor: $N_s=114 \text{ [points]}$ The area per pair of thermocouple: $S_s=S_d \div N_s=0.000625 \div 114=0.00000548 \text{ [m.sup.2]}=S$

(129) Note that the temperature difference ΔT obtained from the area S_d of the heat generating part and the electromotive force V at the heat flow sensor **54**, the number N_s of temperature measurement points of the heat flow sensor **54**, and the area S_s per pair of thermocouples are similar to those in the condition in the calculation example described in the first embodiment.

(130) Each temperature difference for calculating the device temperature T_w is calculated as follows. First, similar to the calculation example in the first embodiment, the vertical temperature difference ΔT_d at the heat generating part temperature measurement point is $8.847[^\circ \text{C}]$ as represented by the following Expression (13).

$$(131) \quad T_d = T_{S_d} = 8.89 \frac{0.000622}{0.000625} = 8.89 \times 0.9952 = 8.847[^\circ \text{C}] \quad \text{Expression(13)}$$

(132) The temperature difference ΔT_c between the temperature measurement points **60** of the heat flow sensor **54** and the holding surface **18A** of the wafer chuck **18** is $4.42[^\circ \text{C}]$ as represented by the following Expression (14).

$$(133) \quad T_c = \frac{Q_d \cdot \text{Math. } L_c}{-K_j \cdot \text{Math. } S_d} = \frac{-K_j \cdot \text{Math. } S_d \cdot \text{Math. } \frac{T_d}{L_j} \cdot \text{Math. } L_c}{-K_j \cdot \text{Math. } S_d} = T_d \frac{L_c}{L_j} = 8.847 \frac{0.005}{0.01} = 4.42[^\circ \text{C}] \quad \text{Expression(14)}$$

(134) The temperature difference ΔT_w between the back of the wafer W and the heat generating part HP is $0.50[^\circ \text{C}]$ as represented by the following Expression (15).

(135)

[Formula15]

$$T_w = \frac{Q_d \cdot \text{Math. } L_w}{-K_w \cdot \text{Math. } S_d} = \frac{-K_j \cdot \text{Math. } S_d \cdot \text{Math. } \frac{T_d}{L_j} \cdot \text{Math. } L_w}{-K_w \cdot \text{Math. } S_d} = T_d \frac{K_j L_w}{K_w L_j} = 8.847 \frac{180 \cdot 0.0005}{160 \cdot 0.01} = 0.5[^\circ \text{C}] \quad \text{Expression(15)}$$

(136) Consequently, the device temperature T_w is $113.77[^\circ \text{C}]$ as represented by the following Expression (16).

[Formula16]

(137)

$$T_w = T_{\text{ref}} + T_d + T_c + T_w = 100 + 8.85 + 4.42 + 0.5 = 113.77[^\circ \text{C}] \quad \text{Expression(16)}$$

(138) The temperature control unit **106A** in the second embodiment calculates the device temperature T_w according to the method of calculating the device temperature described above. The temperature control unit **106A** then adopts the calculated device temperature as the control temperature, and executes temperature control for the wafer chuck **18** so that the device temperature may approach the preliminarily set temperature (e.g., the test temperature).

(139) Consequently, according to the temperature control system **50** in the second embodiment, based on the detection results of the temperature sensors **52**, the detection result of the heat flow sensor **54**, and the device temperature calculation condition data (information about the physical property values, dimensions and the like from the heat generating part temperature measurement point to the heat generating part), the temperature (device temperature) of the device which is the heat generating part of the wafer W, is calculated, and based on the calculated device temperature, the temperature control for the wafer chuck **18** is performed. Accordingly, the electrical property test of the device to be measured may be highly accurately performed at the desired test temperature.

(140) The temperature control system and the temperature control method according to the present invention have thus been described above in detail. However, it is a matter of course that several improvements or modifications may be applied in a range without departing from the gist of the present invention.

REFERENCE SIGNS LIST

(141) **1** . . . Wafer test system, **10** . . . Prober, **18** . . . Wafer chuck, **24** . . . Probe card, **25** . . . Probe, **28** . . . Chuck top suction plate, **34** . . . Thermocouple housing bore, **40** . . . Heating and cooling unit, **42** . . . Cooling plate, **44** . . . Heater, **46** . . . Peltier element, **50, 50A, 50B** . . . Temperature control system, **52** . . . Temperature sensor, **54** . . . Heat flow sensor, **56** . . . Thermocouple, **60** . . . Temperature measurement point, **62** . . . Reference junction, **90** . . . Control device, **92** . . . Cooling device, **94** . . . Heater power source, **96** . . . Peltier power source, **100** . . . Temperature control device, **100A** . . . Temperature control device, **102** . . . Reference temperature detecting unit, **104** . . . Temperature difference detecting unit, **106** . . . Temperature control unit, **106A** . . . Temperature control unit, HP . . . Heat generating part, W . . . Wafer

Claims

1. A wafer chuck having a holding surface for holding a wafer, the wafer chuck comprising: a heating and cooling source configured to heat or cool the wafer chuck; at least one temperature sensor disposed in the wafer chuck; and a heat flow sensor which is disposed in the wafer chuck, and in which a plurality of thermocouples are connected in series and connecting parts between ones adjacent to each other of the thermocouples are alternately disposed at a first depth position from the holding surface and at a second depth position deeper than the first depth position, wherein the temperature sensor is used to measure a reference temperature that is a temperature at the second depth position in a corresponding part being a part corresponding to a heat generating part of the wafer on the wafer chuck, the heat flow sensor has a plurality of temperature measurement points provided for the respective thermocouples, the plurality of temperature measurement points being

disposed over an entirety of the wafer chuck when the wafer chuck is seen in plan view, and the heat flow sensor is used to measure a vertical temperature difference between the first depth position and the second depth position in the corresponding part.

2. The wafer chuck according to claim 1, wherein a plurality of the temperature sensors are disposed in a distributed manner.

3. A temperature control system, comprising: the wafer chuck according to claim 1; and a temperature control unit configured to calculate a temperature of the corresponding part or the heat generating part as a control temperature, based on a detection result of the temperature sensor and on a detection result of the heat flow sensor.

4. The temperature control system according to claim 3, wherein the temperature control unit executes: a process of detecting the reference temperature based on the detection result of the temperature sensor; a process of calculating the vertical temperature difference based on the detection result of the heat flow sensor; and a process of calculating, as the control temperature, a temperature of the heat generating part temperature measurement point that is a temperature of the corresponding part, based on the reference temperature and the vertical temperature difference.

5. The temperature control system according to claim 3, wherein the temperature control unit executes: a process of detecting the reference temperature based on the detection result of the temperature sensor; a process of calculating the vertical temperature difference, based on the detection result of the heat flow sensor; and a process of calculating, as the control temperature, a device temperature that is a temperature of the heat generating part, based on the reference temperature, on the vertical temperature difference, and on data that includes physical property values and dimensions of the wafer and the wafer chuck.

6. The temperature control system according to claim 4, wherein in the wafer chuck, a plurality of the temperature sensors are disposed in a distributed manner, and the temperature control unit detects, as the reference temperature, a lowest temperature among temperatures detected by the plurality of temperature sensors, or an average or a median of the temperatures detected by the plurality of temperature sensors.

7. The temperature control system according to claim 3, wherein the temperature control unit controls the heating and cooling source so that the control temperature may be equal to a preset target temperature.

8. A temperature control method of controlling a temperature of the wafer chuck according to claim 1, the method including: a reference temperature detecting step of detecting, by the at least one temperature sensor, the reference temperature of the wafer chuck; a heat flow detecting step of detecting, by the heat flow sensor, a heat flow caused by partial heat generation of the wafer held on the holding surface; and a temperature control step of calculating the vertical temperature difference, based on a detection result of the heat flow detecting step, and calculating a temperature of the corresponding part or the heat generating part as a control temperature, based on the reference temperature detected in the reference temperature detecting step and the vertical temperature difference.

9. The temperature control method according to claim 8, wherein the temperature control step includes: a process of calculating, as the control temperature, a temperature of a heat generating part temperature measurement point that is a temperature of the corresponding part, based on the reference temperature and the vertical temperature difference.

10. The temperature control method according to claim 8, wherein the temperature control step includes: a process of calculating, as the control temperature, a device temperature that is a temperature of a heat generating part of the wafer, based on the reference temperature, on the vertical temperature difference, and on data that includes physical property values and dimensions of the wafer and the wafer chuck.

11. The temperature control method according to claim 8, wherein the temperature control step

includes a process of heating or cooling the wafer chuck so that the control temperature may be equal to a preset target temperature.
